Malta Fight Co. - Payment Automation System Proposal

To: Jay Rajakariyar, Malta Fight Co.
From: Darshana
Subject: Payment Automation System - Python Application Proposal & Analysis
Date: June 5, 2025

Dear Jay,

Thank you for providing the detailed requirements, sample data files, and comprehensive brief for the Malta Fight Co. payment automation system. After thoroughly analyzing your business needs, I'm writing to present a detailed proposal and explain why a **custom Python desktop application** is the optimal solution for your complex requirements.

Understanding the Complexity

Your payment automation system presents several significant technical challenges that go far beyond a simple spreadsheet solution:

Core Complexities:

- **~50 Different Membership Types** with unique pricing and percentage rules
- **Multi-stakeholder Payment Splits** (Coach/BGM/Management/MFC) with varying percentages
- **Complex Customer Matching** between GoTeamUp attendance and historical payment data
- **Advanced Business Logic** for private vs. group sessions with different calculation rules
- **Extensive Exception Handling** for refunds, discounts, free sessions, parent/child profiles
- **Multi-pack Calculations** requiring session distribution over time periods
- **Tax Integration** (7% deduction) and discount processing
- **Data Validation** across multiple CSV exports with potential inconsistencies

Data Processing Challenges:

- **Customer Name Matching** across different systems with potential variations
- **Package Assignment Logic** ensuring latest purchased package is used

- **Drop-in Session Handling** with different payment structures
- **Historical Data Migration** while maintaining calculation accuracy
- **Scalability Requirements** as your business grows and adds new package types

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Why Google Sheets/Excel Solutions Are Inadequate

While spreadsheets might seem like a simpler approach, they present critical limitations for your requirements:

Technical Limitations:

- **Formula Complexity:** Your business logic would require hundreds of nested IF statements and complex VLOOKUP chains that become unmaintainable
- **Performance Issues:** Processing 600+ attendance records against 667+ payment records monthly will cause significant slowdowns
- **Error Prone:** Manual formula construction for 50+ membership types creates high risk of calculation errors
- **Data Integrity:** No built-in validation for customer matching or exception handling
- **Scalability Problems:** As your business grows, spreadsheet performance degrades exponentially

Business Risk Factors:

- **Formula Breakage:** Users accidentally modifying formulas leading to incorrect payments
- **Version Control:** Multiple people editing creates conflicts and data loss risks
- **Audit Trail:** Limited ability to track changes and verify calculation accuracy
- **Exception Handling:** Manual workarounds required for complex business scenarios
- **Future Modifications:** Adding new membership types requires extensive formula updates

User Experience Issues:

- **Steep Learning Curve:** Staff need advanced Excel knowledge to maintain the system
- **Error Debugging:** Identifying calculation errors in complex formulas is extremely difficult
- **Report Generation:** Limited professional output capabilities for payslips and summaries

- **Data Import:** Manual copy-paste processes prone to human error

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Proposed Solution: Custom Python Desktop Application

I recommend developing a **professional Windows desktop application** using Python and PyQt that addresses all your requirements with enterprise-level reliability.

Why Python Desktop Application?

Technical Advantages:

- **Robust Data Processing:** Pandas library handles large datasets efficiently with advanced matching algorithms
- **Complex Business Logic:** Clean, maintainable code structure for your intricate payment rules
- **Error Handling:** Comprehensive validation and exception management built-in
- **Performance:** Fast processing of large data sets without performance degradation
- **Scalability:** Easy to add new membership types and business rules without system redesign

Business Benefits:

- **User-Friendly Interface:** Intuitive dashboard requiring no technical knowledge
- **Automated Processing:** One-click monthly payment calculations eliminating manual work
- **Professional Reports:** Generate polished payslips and summaries in PDF/Excel formats
- **Data Integrity:** Built-in validation prevents calculation errors and data corruption
- **Audit Trail:** Complete tracking of all calculations and modifications
- **Future-Proof:** Easy to modify and extend as your business evolves

Operational Advantages:

- **Standalone Application:** No internet dependency or cloud security concerns
- **Windows Optimization:** Designed specifically for Windows environment
- **Local Data Storage:** Complete control over sensitive payment information

- **Backup Integration:** Automated data backup capabilities
- **Multi-User Ready:** Can be expanded for multiple users if needed

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Application Features & Functionality

Core Features (Included in $1,500 Budget):

1. Data Import Manager

- Automated CSV import from GoTeamUp exports
- Data validation and error detection
- Preview and verification before processing
- Historical data integration

2. Intelligent Matching Engine

- Advanced customer name matching algorithm
- Latest package assignment logic
- Duplicate profile handling
- Exception flagging for manual review

3. Payment Calculation Engine

- Automated percentage splits for all stakeholders
- Private vs. group session differentiation
- Multi-pack session distribution
- Tax calculation and discount processing
- Refund and free session handling

4. Rule Management System

- Easy interface to add/modify membership types
- Percentage configuration for each stakeholder
- Special rules and exceptions handling
- Visual rule validation

5. Dashboard & Reporting

- Monthly overview with key metrics
- Payment summaries for coaches, BGM, and management
- CSV export capabilities for further analysis
- Historical report access

6. Settings Management

- Coach information management
- Company settings configuration
- System backup and maintenance tools

System Requirements:

- **Operating System:** Windows 10/11 (64-bit)
- **Memory:** 4GB RAM minimum, 8GB recommended
- **Storage:** 2GB available space
- **Processor:** Intel i3 or equivalent

Prototype Demonstration

I have developed a comprehensive **working prototype** that demonstrates the complete user interface and workflow. **Please find the video demonstration attached** showing:

- **Dashboard Overview** with real-time statistics and dark theme functionality
- **Data Import Process** for both attendance and payment data
- **Rule Management Interface** for configuring membership types and percentages
- **Payment Calculator** with stakeholder breakdowns
- **Report Generation** capabilities
- **Settings Management** for system configuration

The prototype uses your actual data structure and business rules, giving you a clear vision of the final application's capabilities.

Investment & Timeline

Core Application Development

Total Investment: $1,500 USD
Development Timeline: 6-8 weeks
Support Period: 30 days post-delivery for adjustments and minor modifications

What's Included:

✅ Complete functional application as demonstrated in prototype
✅ Data import and processing capabilities
✅ Payment calculation engine with all business rules
✅ User-friendly interface with dashboard and reporting
✅ Rule management system for future modifications
✅ Installation package and user documentation
✅ Basic training session via video call
✅ 30 days of support for bug fixes and minor adjustments

Future Enhancement Options (Optional):

- **Advanced PDF Payslip Generation:** +$400
- **Email Integration for Report Distribution:** +$300
- **Advanced Exception Handling Interface:** +$350
- **Multi-User Access Control:** +$500
- **Cloud Backup Integration:** +$250

Why This Investment Makes Sense

Cost-Benefit Analysis:

- **Time Savings:** 20+ hours monthly automation = $500+ in labor savings (at $25/hour)
- **Error Reduction:** Eliminate costly payment calculation mistakes
- **Professional Image:** Generate professional payslips and reports
- **Scalability:** Handle business growth without system limitations
- **Peace of Mind:** Reliable, auditable payment processing

Return on Investment:

- **Payback Period:** ~3 months based on time savings alone
- **Long-term Value:** 5+ years of reliable operation
- **Business Growth Support:** No need for system replacement as you expand

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Next Steps

If you're satisfied with the prototype demonstration and this proposal:

1. **Confirm Project Approval** and preferred timeline
2. **Provide Final Data Samples** for testing and validation
3. **Schedule Kick-off Meeting** to finalize requirements
4. **Begin Development** with weekly progress updates
5. **User Acceptance Testing** with your team
6. **Deployment and Training** at your facility

I'm confident this solution will transform your payment processing workflow, eliminate manual errors, and provide the professional automation system Malta Fight Co. deserves.

Please review the attached prototype video and let me know if you have any questions or would like to discuss any aspects of this proposal.

Looking forward to working with you on this exciting project!

Best regards,

Darshana
Data Analytics Specialist
14+ Years Experience | 565+ Completed Projects
Specialized in Advanced Automation Solutions

P.S. The prototype video clearly demonstrates why a custom application is superior to spreadsheet solutions – please pay attention to the user interface design, data processing capabilities, and professional reporting features that would be impossible to replicate in Excel or Google Sheets.